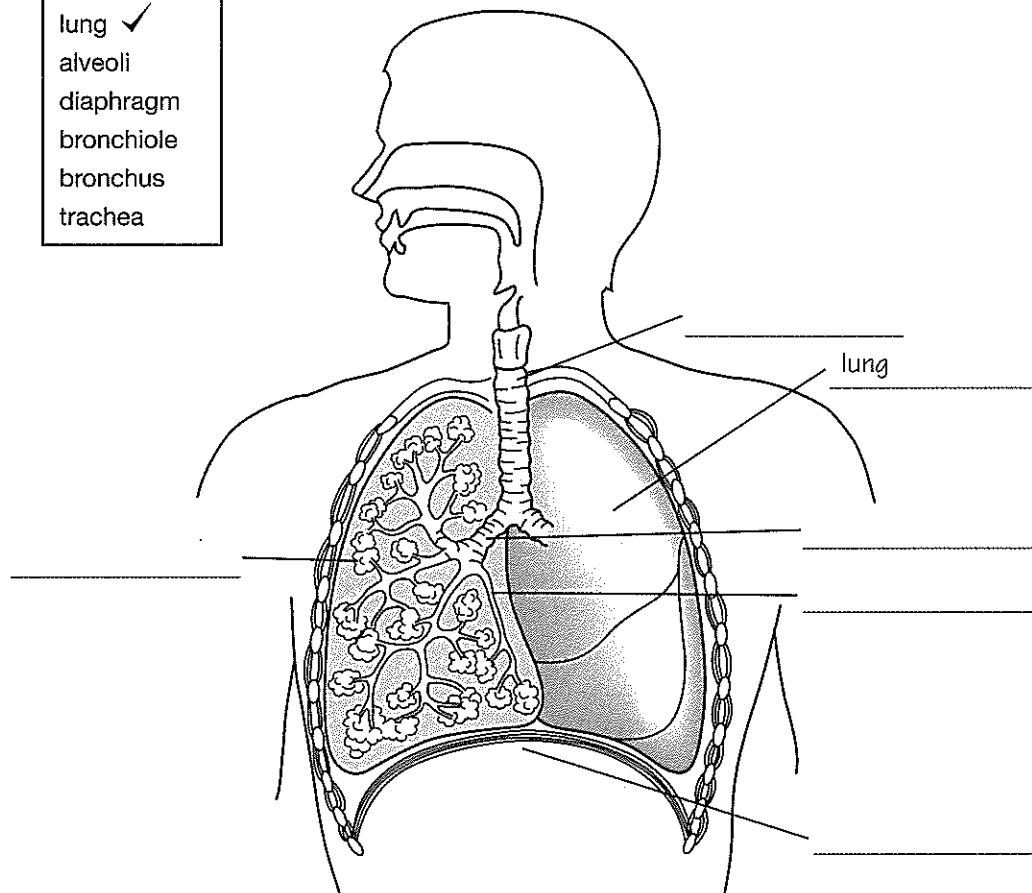


Science understanding

Verbal/Linguistic Visual/Spatial

- 1 Select terms from the list below to **name** the parts of the respiratory system indicated.

lung ✓
alveoli
diaphragm
bronchiole
bronchus
trachea



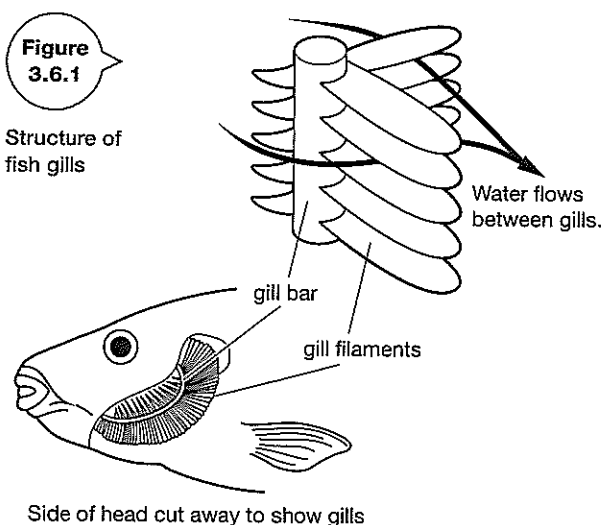
- 2 Draw lines to match the parts of the respiratory system with their description.

Part of the respiratory system	Description
Trachea	A sheet of muscle that separates the chest from the abdomen. It contracts and flattens as you breathe in and arches up as you breathe out.
Bronchi	A cluster of sacs in which oxygen and carbon dioxide are exchanged.
Alveoli	Thin-walled tube reinforced with rings of cartilage. You can feel these rings as ridges on the front of your throat.
Diaphragm	One of these carries air into each lung.

Science understanding, Science inquiry

Verbal/Linguistic Visual/Spatial

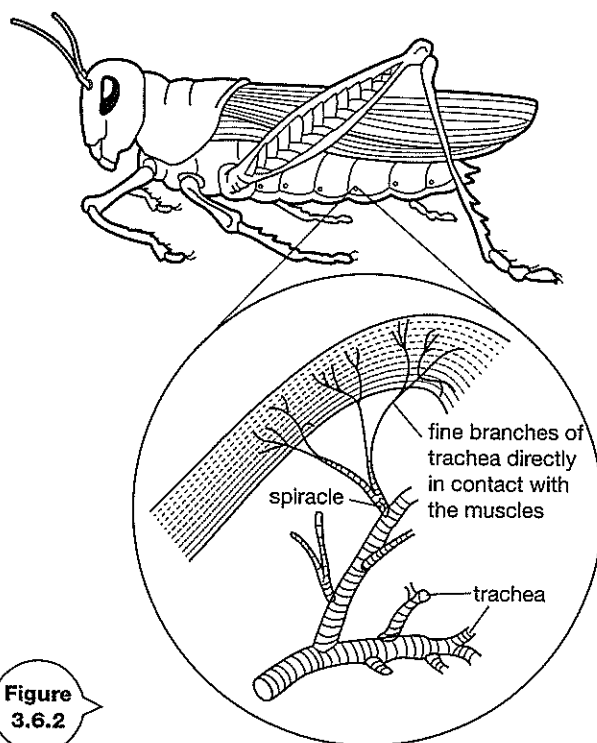
In humans, millions of tiny alveoli provide a large surface area through which the gases oxygen and carbon dioxide can be exchanged. Other animals have slightly different respiratory systems. Fish have gills, insects have structures called trachea and earthworms exchange gases through their skin. You are going to compare these respiratory surfaces.



The gills of fish have a large surface area. Gills are made up of many fine filaments (thread-like structures), each of which has a very good blood supply.

Gills are always moist because they are surrounded by water.

moist (adj)
a bit wet; damp



Insects have small tubes called trachea that carry air to every cell of the body. The ends of the small branches of the trachea are moist. Oxygen dissolves at the moist surface and then moves into the cells.

Small openings called spiracles on the outside of the insect's body close the trachea when the insect is not active. This prevents water being lost and makes sure that the ends of the trachea do not dry out.

prevent (v)
to stop something happening

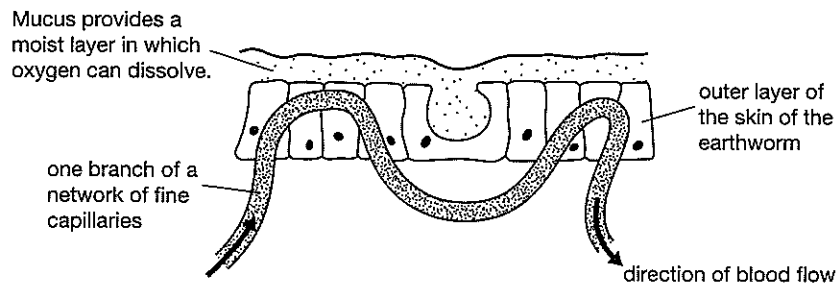


Figure 3.6.3

An earthworm's skin acts as a gas exchange surface.

Earthworms produce a layer of mucus over their skin. Mucus is a thin jelly that keeps the skin moist. Oxygen from the air dissolves at the moist surface. The earthworm has many blood capillaries close to the surface and its blood contains haemoglobin and is red, just like human blood. The haemoglobin collects the dissolved oxygen and the blood carries it to the cells.

capillaries (n) very small blood vessels

haemoglobin (n) a chemical in the blood that carries oxygen around the body

- 1 What characteristics make an efficient gas exchange surface? **List** them.

Efficient gas exchange surfaces have:

- _____
- _____
- _____



- 2 **Compare** the gas exchange surfaces of humans, fish, insects and worms. Use the characteristics you have listed in question 1.

- 3 The maximum amounts of oxygen that can be supplied to each gram of muscle tissue in 1 hour are: earthworm 60 mm³, mouse running 20 000 mm³ and butterfly flying 100 000 mm³.

- (a) **Explain** how a butterfly is able to supply oxygen to its muscles more quickly than a mouse can.

- (b) **Explain** why the earthworm has such a limited supply of oxygen.

limited (adj) a small amount; restricted

- (c) Where do you think a human would fit in the list? **Justify** your response.
